BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-049662
Article Type:	Protocol
Date Submitted by the Author:	02-Feb-2021
Complete List of Authors:	Mohamed-Yassin, Mohamed-Syarif; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Baharudin, Noorhida; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Abdul-Razak, Suraya; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Cardiac Vascular and Lungs Research Institute (CaVaLRI) Ramli, Anis Safura; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM) Lai, Nai Ming; Taylor's University, School of Medicine
Keywords:	Lipid disorders < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY, Cardiology < INTERNAL MEDICINE

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Mohamed-Syarif Mohamed-Yassin[1§], Noorhida Baharudin[1], Suraya Abdul-Razak[1,2], Anis Safura Ramli[1,3], Lai Nai-Ming[4]

[1] Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Jalan Prima Selayang 7, 68100 Batu Caves, Selangor, MALAYSIA

[2] Cardiac Vascular and Lungs Research Institute (CaVaLRI), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[3] Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[4] School of Medicine, Taylor's University, 1, Jalan Taylor's, 47500 Subang Jaya, Selangor, MALAYSIA

[§]**Corresponding Author**:

Name: Dr Mohamed-Syarif Mohamed-Yassin

Postal Address: Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Selayang Campus, Jalan Prima Selayang 7, Batu Caves, 68100 Selangor, MALAYSIA.

Email: syarif8258@uitm.edu.my

Contact No.: +6012 970 2295

For peer review only

Authors' details and e-mail address:

Author 1: MSMY

Given names: Mohamed-Syarif

Family name: Mohamed-Yassin

Preferred abbreviated names: Mohamed-Yassin MS

Email: syarif8258@uitm.edu.my

Author 2: NB

Given names: Noorhida

Family name: Baharudin

Preferred abbreviated names: Baharudin N

Email: noorhida8229@uitm.edu.my

Author 3: SAR

Given names: Suraya

Family name: Abdul-Razak

Email: suraya617@uitm.edu.my

Preferred abbreviated names: Abdul-Razak S

For peer review only

Author 4: ASR

Given names: Anis Safura

Family name: Ramli

Email: anis014@uitm.edu.my

Preferred abbreviated names: Ramli AS

Author 5: LNM

Given names: Nai Ming

Family name: Lai

Email: naiming.lai@taylors.edu.my

Preferred abbreviated names: Lai NM

Word count and references:

- Abstract: 300
- Main text: 1341
- References: 23

For peer review only

ABSTRACT

Introduction: Dyslipidaemia is one of the established risk factors for cardiovascular disease. Evidence from large-scale trials showed that effective treatment of dyslipidaemia can reduce all-cause and coronary mortality. To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults. We propose to perform a systematic review on the global prevalence of dyslipidaemia in adults 18 years and older.

Methods and analysis: We will identify observational studies through comprehensive literature searches. We will search: MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov. Two reviewers will independently screen the titles and abstracts, attain full text of eligible articles, extract data, and appraise the quality and bias of the included studies. Disagreement among the authors will be resolved by discussion leading to a consensus. Next, we will perform a narrative synthesis of the study results. Study heterogeneity will be assessed using I^2 statistics. If I^2 is high (≥75%), and plausible heterogeneity contributors are found, we will divide the studies into appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression. If I^2 <75%, we will undertake meta-analysis using the random-effects model and transform all prevalence estimates using the Freeman-Tukey transformation for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence. We will then back-transform the point estimate, and report our results using the back-transformed figures.

Ethics and dissemination: Ethics approval is not a requirement as this study is based on available published data. Results of this systematic review will be presented at conferences, shared with relevant health authorities, and published in a peer-reviewed journal. These results may help quantify the magnitude of dyslipidaemia globally, and guide preventative and therapeutic interventions.

Trial registration: PROSPERO registration number - CRD42020200281

ARTICLE SUMMARY

Strengths and limitations of this study

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

- This is the first systematic review on the global prevalence of dyslipidaemia in adult populations
- The overall pooled prevalence of dyslipidaemia, as well as the prevalence in different geographical regions and settings will help provide the baseline risk and absolute risk difference in studies assessing interventions for dyslipidaemia.
- Possible limitation of this systematic review could be the lack of studies and data on the subtypes of dyslipidaemia.

INTRODUCTION

Dyslipidaemia refers to lipid abnormalities consisting of either one or any combination of the following: elevated total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), or low high-density lipoprotein cholesterol (HDL-c).[1] Dyslipidaemia is one of the established risk factors for cardiovascular disease.[1] Extensive reviews had concluded that elevated LDL-c is a significant contributor to ASCVD.[2-4] Low HDL-c had also been found to be associated with CVD.[5-8] However, recent studies had questioned the role of high TC and LDL-c, in the development of atherosclerosis and CVD.[9 10] The protective role of increasing HDL-c in CVD had also been challenged[11], while some studies had shown that non-HDL-c predicts CV risk better than LDL-c.[12-14]

The World Health Organization's estimates from 2008 showed that the prevalence of hypercholesterolaemia were the highest in Europe (53.7%) and America (47.7%), while South East Asia (30.3%) and Africa (23.1%) had much lower prevalence.[15] However, Lin et al. reported marked prevalence differences between different Asia Pacific countries, ranging from 9% in Indonesia to 46.9% in the Philippines.[16] For high LDL-c, high TG, and low HDL-c, the prevalence ranges from 7.8 to 47.2%, 13.9 to 38.6%, and 10.1 to 71.3%, respectively.[16] Noubiap et al. in their 2018 systematic review of dyslipidaemia in Africa, reported the prevalence of elevated TC, elevated LDL-c, elevated TG, and low HDL-c as 25.5%, 28.6%, 17%, and 37.4%, respectively.[17]

Vigorous evidence from large-scale randomised trials showed that effective treatment of dyslipidaemia can reduce all-cause mortality and coronary mortality.[18] LDL-c reduction of 1 mmol/L with statin treatment can lower the 5-year incidence of stroke, coronary revascularisation, and major coronary events by around one fifth.[2 18]

To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults and we aim to address this evidence gap. Hence, the objective of this systematic review is to determine the global prevalence of dyslipidaemia in adults 18 years old and older.

Here, we present the protocol for this systematic review, prepared according to the PRISMA-P 2015 guidelines.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS AND ANALYSIS

Patient and public involvement

No patient involved.

Study design

This will be a systematic review with meta-analysis if suitable data are available for pooling.

Searches

We will search MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov for ongoing studies. There will be no restrictions applied on the publication period and language.

Search strategy

Our MEDLINE search strategy is shown as follow:

#14 Add Search (#10 AND #13)

#13 Add Search (#11 OR #12)

#12 Add Search prevalence[MeSH Terms]

#11 Add Search prevalence[Title/Abstract]

#10 Add Search (#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9)

#9 Add Search “lipid disorder”[Title/Abstract]

#8 Add Search hypertriglyceridemia[MeSH Terms]

#7 Add Search hypertriglycerid*[Title/Abstract]

#6 Add Search essential hypercholesterolemia[MeSH Terms]

#5 Add Search hypercholesterol*[Title/Abstract]

#4 Add Search hyperlipidemia[MeSH Terms]

#3 Add Search hyperlipid*[Title/Abstract]

#2 Add Search dyslipidemia [MeSH Terms]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

#1 Add Search dyslipid*[Title/Abstract]

Participants/Population

This systematic review will include adults 18 years or older with dyslipidaemia (elevated total cholesterol, elevated LDL-c, elevated TG, low HDL-c).

Types of study to be included

Observational studies (prevalence studies, which can be in the form of a cross-sectional or longitudinal observational study) will be included.

Data extraction (selection and coding)

We will include all cross-sectional and longitudinal observational studies reporting on the prevalence of elevated serum total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), and low high-density lipoprotein cholesterol (HDL-c) in adults above 18 years old. We will exclude studies of children or individuals with familial hypercholesterolaemia in this review. We will also exclude editorials, commentaries, reviews, letters, case series with less than 50 patients, and studies without primary data or explicit description of methods. When a study is published in two or more reports, the most comprehensive one with the largest sample size will be included. Two review authors will independently screen the titles and abstracts of the articles retrieved from the literature search. These citations will be imported into the Endnote software, with duplicate articles being omitted. Any studies excluded from the review will be documented with justification. Full texts of potentially eligible articles will be attained and further evaluated for final inclusion. Any disagreement among the review authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary. We will seek professional translation services for abstracts and full articles that are not available in English.

Two review authors will independently extract relevant data from individual studies using a pro forma designed specifically for this review. The following data will be extracted: first author's name, publication year, study design, country, locality (rural vs urban), setting (community or hospital-based), sample size, mean/median age, age range, proportion of men/women, any disease specific to the study population, dyslipidaemia subtypes included (ie elevated TC, elevated LDL-c, elevated TG, low HDL-c), diagnostic cut-off levels, and the number of participants with dyslipidaemia. We will

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

consider the most frequently used cut-off for the diagnosis of dyslipidaemia. Any disagreement among the review authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary.

Risk of bias (quality) assessment

We will evaluate all included studies for quality and bias using an adapted version of the Risk of Bias Tool for Prevalence Studies developed by Hoy et al.[19] or the Joanna Briggs Institute's Checklist for Prevalence Studies[20], pending team deliberation and piloting if required.

Strategy for data synthesis

First, we will perform a narrative (descriptive) synthesis of the study results. We will assess the degree of heterogeneity in the estimates among studies as measured by the I square (I^2) statistics, which denotes the percentage of variation in estimate that is not attributable to sampling error and will be relevant only if there are more than one included study in the analysis. We adopt an I^2 level of 75% to indicate a substantial degree of heterogeneity.[21] As heterogeneity is inherent among prevalence studies, we adopt a higher threshold of I^2 estimate compared to that in a systematic review of randomized studies.

If I^2 of the pool of studies is below 75%, we will undertake meta-analysis using the random-effects model, to take into account of the inherent heterogeneity present among different prevalent studies. To address the limitations of current meta-analysis of prevalence studies, in which studies with very low or very high prevalence tends to be assigned disproportionately high weight due to the marked reduction of their inferred variance towards 0, we will transform all prevalence estimates using the Freeman-Tukey transformation (arcsine square root transformation)[22] for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence interval[23] , using the MetaXL software as an add-in of Excel (EpiGear International, QLD, Australia). We will then back-transform the point estimate and 95% confidence interval, and report our results using the back-transformed figures to facilitate interpretation.

If the I^2 is 75% or above, which indicates substantial level of heterogeneity, we will explore possible contributors of heterogeneity by assessing variation between studies in terms of the following: i. Population characteristics, including geographical location, age group and sex, ii. Definition and measurement of the target condition of dyslipidaemia, including the data collection methods (e.g. self-completed questionnaire vs. interview or testing (clinical assessment and/or objective test), iii. Risk of bias of the included studies, divided by low or high risks of bias in participation and outcome

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

measurement. If we find plausible contributors of heterogeneity, we will divide the studies into appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression (Stata 13 software).

Sensitivity analysis

If sufficient data are available, we will perform sensitivity analysis to assess the impact of excluding studies with an overall high risk of bias.

Certainty of evidence

As there is not yet an established tool to rate the certainty of evidence for prevalence studies, we will discuss our findings by incorporating our assessment of the risk of bias of included studies and our findings in sensitivity analysis.

ETHICS AND DISSEMINATION

As the current study is based on available published data, ethics approval is not a requirement. We plan to disseminate the findings from this systematic review at relevant international conferences. Then, we will submit the final report in the form of an article to a peer-reviewed journal for publication. We also plan to share the findings with the relevant health authorities such as the World Health Organization and guideline development bodies. Finally, we plan to update the review in the future as more relevant publications are produced.

Author Contributions MSMY and LNM conceived and designed the protocol. MSMY drafted the manuscript. NB, SAR, ASR, and LNM revised the manuscript for methodological and clinical content. All authors read and approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors

Conflict of interest None

Patient consent for publication Not applicable.

REFERENCES

1. Health Technology Assessment Unit. Clinical Practice Guidelines on Management of Dyslipidaemia: Ministry of Health Malaysia, 2017:105.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2. Collins R, Reith C, Emberson J, et al. Interpretation of the evidence for the efficacy and safety of statin therapy. *Lancet* 2016;388(10059):2532-61. doi: 10.1016/S0140-6736(16)31357-5 [published Online First: 2016/09/13]

3. Ference BA, Ginsberg HN, Graham I, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J* 2017;38(32):2459-72. doi: 10.1093/eurheartj/ehx144 [published Online First: 2017/04/27]

4. Silverman MG, Ference BA, Im K, et al. Association Between Lowering LDL-C and Cardiovascular Risk Reduction Among Different Therapeutic Interventions: A Systematic Review and Meta-analysis. *JAMA* 2016;316(12):1289-97. doi: 10.1001/jama.2016.13985 [published Online First: 2016/09/28]

5. Emerging Risk Factors Collab., Di Angelantonio E, Sarwar N, et al. Major lipids, apolipoproteins, and risk of vascular disease. *JAMA* 2009;302(18):1993-2000. doi: 10.1001/jama.2009.1619 [published Online First: 2009/11/12]

6. Gordon DJ, Probstfield JL, Garrison RJ, et al. High-density lipoprotein cholesterol and cardiovascular disease. Four prospective American studies. *Circulation* 1989;79(1):8-15. doi: 10.1161/01.cir.79.1.8 [published Online First: 1989/01/01]

7. Hubert HB, Feinleib M, McNamara PM, et al. Obesity as an independent risk factor for cardiovascular disease: a 26-year follow-up of participants in the Framingham Heart Study. *Circulation* 1983;67(5):968-77. doi: 10.1161/01.cir.67.5.968 [published Online First: 1983/05/01]

8. Rosenson RS. Low HDL-C: a secondary target of dyslipidemia therapy. *Am J Med* 2005;118(10):1067-77. doi: 10.1016/j.amjmed.2004.12.021 [published Online First: 2005/10/01]

9. Hamazaki T, Okuyama H, Ogushi Y, et al. Towards a Paradigm Shift in Cholesterol Treatment. A Re-examination of the Cholesterol Issue in Japan. *Ann Nutr Metab* 2015;66 Suppl 4:1-116. doi: 10.1159/000381654 [published Online First: 2015/05/01]

10. Ravnskov U, de Lorgeril M, Diamond DM, et al. LDL-C does not cause cardiovascular disease: a comprehensive review of the current literature. *Expert Rev Clin Pharmacol* 2018;11(10):959-70. doi: 10.1080/17512433.2018.1519391 [published Online First: 2018/09/11]

11. Voight BF, Peloso GM, Orho-Melander M, et al. Plasma HDL cholesterol and risk of myocardial infarction: a mendelian randomisation study. *Lancet* 2012;380(9841):572-80. doi: 10.1016/S0140-6736(12)60312-2 [published Online First: 2012/05/23]

12. Arsenault BJ, Rana JS, Stroes ES, et al. Beyond low-density lipoprotein cholesterol: respective contributions of non-high-density lipoprotein cholesterol levels, triglycerides, and the total cholesterol/high-density lipoprotein cholesterol ratio to coronary heart disease risk in apparently healthy men and women. *J Am Coll Cardiol* 2009;55(1):35-41. doi: 10.1016/j.jacc.2009.07.057 [published Online First: 2010/02/02]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

13. Ramjee V, Sperling LS, Jacobson TA. Non-high-density lipoprotein cholesterol versus apolipoprotein B in cardiovascular risk stratification: do the math. *J Am Coll Cardiol* 2011;58(5):457-63. doi: 10.1016/j.jacc.2011.05.009 [published Online First: 2011/07/23]
14. Robinson JG, Wang S, Smith BJ, et al. Meta-analysis of the relationship between non-high-density lipoprotein cholesterol reduction and coronary heart disease risk. *J Am Coll Cardiol* 2009;53(4):316-22. doi: 10.1016/j.jacc.2008.10.024 [published Online First: 2009/01/24]
15. World Health Organization. Global health observatory data repository Geneva: World Health Organization; 2016 [cited 2020 4 January]. Available from: https://apps.who.int/gho/data/view.main.2467?lang=en accessed 4 January 2020.
16. Lin CF, Chang YH, Chien SC, et al. Epidemiology of Dyslipidemia in the Asia Pacific Region. *Int J Gerontol* 2018;12(1):2-6. doi: 10.1016/j.ijge.2018.02.010
17. Noubiap JJ, Bigna JJ, Nansseu JR, et al. Prevalence of dyslipidaemia among adults in Africa: a systematic review and meta-analysis. *Lancet Glob Health* 2018;6(9):e998-e1007. doi: 10.1016/s2214-109x(18)30275-4 [published Online First: 2018/08/15]
18. Baigent C, Keech A, Kearney PM, et al. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;366(9493):1267-78. doi: 10.1016/S0140-6736(05)67394-1 [published Online First: 2005/10/11]
19. Hoy D, Brooks P, Woolf A, et al. Assessing risk of bias in prevalence studies: modification of an existing tool and evidence of interrater agreement. *J Clin Epidemiol* 2012;65(9):934-9. doi: 10.1016/j.jclinepi.2011.11.014 [published Online First: 2012/06/30]
20. Munn Z, Moola S, Lisy K, et al. Methodological guidance for systematic reviews of observational epidemiological studies reporting prevalence and cumulative incidence data. *Int J Evid Based Healthc* 2015;13(3):147-53. doi: 10.1097/XEB.0000000000000054 [published Online First: 2015/09/01]
21. Barendregt JJ, Doi SA, Lee YY, et al. Meta-analysis of prevalence. *J Epidemiol Community Health* 2013;67(11):974-8. doi: 10.1136/jech-2013-203104 [published Online First: 2013/08/22]
22. Freeman MF, Tukey JW. Transformations Related to the Angular and the Square Root. *Ann Math Statist* 1950;21(4):607-11. doi: 10.1214/aoms/1177729756
23. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P 2015 Checklist

This checklist has been adapted for use from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	94
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	3–55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	249-251
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	N/A
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐	☒	N/A
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	☒	N/A
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☒	N/A
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	130-131
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	131-132
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	167-183

For peer review only

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	143-146
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	149-164
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	175-198
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	183-189
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	190-191
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	167-198
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	176-179
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	200-203
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	213-222
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	206-212
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	229-234
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	206
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	N/A
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	235-238

For peer review only

BMJ Open

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-049662.R1
Article Type:	Protocol
Date Submitted by the Author:	21-Oct-2021
Complete List of Authors:	Mohamed-Yassin, Mohamed-Syarif; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Baharudin, Noorhida; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Abdul-Razak, Suraya; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Cardiac Vascular and Lungs Research Institute (CaVaLRI) Ramli, Anis Safura; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM) Lai, Nai Ming; Taylor's University, School of Medicine
Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Cardiovascular medicine, Global health
Keywords:	Lipid disorders < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY, Cardiology < INTERNAL MEDICINE

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Mohamed-Syarif Mohamed-Yassin[1§], Noorhida Baharudin[1], Suraya Abdul-Razak[1,2], Anis Safura Ramli[1,3], Lai Nai-Ming[4]

[1] Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Jalan Prima Selayang 7, 68100 Batu Caves, Selangor, MALAYSIA

[2] Cardiac Vascular and Lungs Research Institute (CaVaLRI), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[3] Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[4] School of Medicine, Taylor's University, 1, Jalan Taylor's, 47500 Subang Jaya, Selangor, MALAYSIA

[§]**Corresponding Author**:

Name: Dr Mohamed-Syarif Mohamed-Yassin

Postal Address: Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Selayang Campus, Jalan Prima Selayang 7, Batu Caves, 68100 Selangor, MALAYSIA.

Email: syarif8258@uitm.edu.my

Contact No.: +6012 970 2295

For peer review only

Authors' details and e-mail address:

Author 1: MSMY

Given names: Mohamed-Syarif

Family name: Mohamed-Yassin

Preferred abbreviated names: Mohamed-Yassin MS

Email: syarif8258@uitm.edu.my

Author 2: NB

Given names: Noorhida

Family name: Baharudin

Preferred abbreviated names: Baharudin N

Email: noorhida8229@uitm.edu.my

Author 3: SAR

Given names: Suraya

Family name: Abdul-Razak

Email: suraya617@uitm.edu.my

Preferred abbreviated names: Abdul-Razak S

For peer review only

Author 4: ASR

Given names: Anis Safura

Family name: Ramli

Email: anis014@uitm.edu.my

Preferred abbreviated names: Ramli AS

Author 5: LNM

Given names: Nai Ming

Family name: Lai

Email: naiming.lai@taylors.edu.my

Preferred abbreviated names: Lai NM

Word count and references:

- Abstract: 300
- Main text: 1471
- References: 23
- Table: 1

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

ABSTRACT

Introduction: Dyslipidaemia is one of the established risk factors for cardiovascular disease. Evidence from large-scale trials showed that effective treatment of dyslipidaemia can reduce all-cause and coronary mortality. To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults. We propose to perform a systematic review on the global prevalence of dyslipidaemia in adults 18 years and older.

Methods and analysis: We will identify observational studies through comprehensive literature searches. We will search: MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov. Two reviewers will independently screen the titles and abstracts, attain full text of eligible articles, extract data, and appraise the quality and bias of the included studies. Disagreement among the authors will be resolved by discussion leading to a consensus. Next, we will perform a narrative synthesis of the study results. Study heterogeneity will be assessed using I^2 statistics. If I^2 is high (≥75%), and plausible heterogeneity contributors are found, we will divide the studies into appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression. If I^2 <75%, we will undertake meta-analysis using the random-effects model and transform all prevalence estimates using the Freeman-Tukey transformation for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence. We will then back-transform the point estimate, and report our results using the back-transformed figures.

Ethics and dissemination: Ethics approval is not a requirement as this study is based on available published data. Results of this systematic review will be presented at conferences, shared with relevant health authorities, and published in a peer-reviewed journal. These results may help quantify the magnitude of dyslipidaemia globally, and guide preventative and therapeutic interventions.

Trial registration: PROSPERO registration number - CRD42020200281

ARTICLE SUMMARY

Strengths and limitations of this study

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

- This is the first systematic review on the global prevalence of dyslipidaemia in adult populations
- The overall pooled prevalence of dyslipidaemia, as well as the prevalence in different geographical regions and settings will help provide the baseline risk and absolute risk difference in studies assessing interventions for dyslipidaemia.
- Possible limitation of this systematic review could be the lack of studies and data on the subtypes of dyslipidaemia.

INTRODUCTION

Dyslipidaemia refers to lipid abnormalities consisting of either one or any combination of the following: elevated total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), or low high-density lipoprotein cholesterol (HDL-c).[1] Dyslipidaemia is one of the established risk factors for cardiovascular disease.[1] Extensive reviews had concluded that elevated LDL-c is a significant contributor to atherosclerotic cardiovascular disease (ASCVD).[2-4] Low HDL-c had also been found to be associated with CVD.[5-8] However, recent studies had questioned the role of high TC and LDL-c, in the development of atherosclerosis and CVD.[9 10] The protective role of increasing HDL-c in CVD had also been challenged[11], while some studies had shown that non-HDL-c predicts CV risk better than LDL-c.[12-14]

The World Health Organization's estimates from 2008 showed that the prevalence of hypercholesterolaemia in adults were the highest in Europe (53.7%) and America (47.7%), while South East Asia (30.3%) and Africa (23.1%) had much lower prevalence.[15] However, Lin et al. reported marked prevalence differences between different Asia Pacific countries, ranging from 9% in Indonesia to 46.9% in the Philippines.[16] For high LDL-c, high TG, and low HDL-c, the prevalence ranges from 7.8 to 47.2%, 13.9 to 38.6%, and 10.1 to 71.3%, respectively.[16] Noubiap et al. in their 2018 systematic review of dyslipidaemia in Africa, reported the prevalence of elevated TC, elevated LDL-c, elevated TG, and low HDL-c as 25.5%, 28.6%, 17%, and 37.4%, respectively.[17]

Vigorous evidence from large-scale randomised trials showed that effective treatment of dyslipidaemia can reduce all-cause mortality and coronary mortality.[18] LDL-c reduction of 1 mmol/L with statin treatment can lower the 5-year incidence of stroke, coronary revascularisation, and major coronary events by around one fifth.[2 18]

To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults and we aim to address this evidence gap. Hence, the objective of this systematic review is to determine the global prevalence of dyslipidaemia in adults 18 years old and older.

Here, we present the protocol for this systematic review, prepared according to the PRISMA-P 2015 guidelines.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS AND ANALYSIS

Actual start date of study

1st September 2020.

Anticipated end date of study

28th February 2022.

Patient and public involvement

No patient involved.

Study design

This will be a systematic review with meta-analysis if suitable data are available for pooling.

Search strategy

We will search PubMed/MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov for ongoing studies. There will be no restrictions applied on the publication period and language. The main search strategies according to the different databases are shown in Table 1.

Table 1 Search strategies

A. Search strategy PubMed/MEDLINE	
Search	Search terms
1	dyslipid*[Title/Abstract]
2	dyslipidemia[MeSH Terms]
3	hyperlipid*[Title/Abstract]
4	hyperlipidemia[MeSH Terms]
5	hypercholesterol*[Title/Abstract]
6	essential hypercholesterolemia[MeSH Terms]
7	hypertriglycerid*[Title/Abstract]
8	hypertriglyceridemia[MeSH Terms]
9	“lipid disorder”[Title/Abstract]
10	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9
11	prevalence[Title/Abstract]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12	prevalence[MeSH Terms]
13	#11 OR #12
14	#10 AND #13
B. Search strategy CENTRAL	
Search	Search terms
1	dyslipid*[Title/Abstract]
2	dyslipidemia[MeSH Terms]
3	hyperlipid*[Title/Abstract]
4	hyperlipidemia[MeSH Terms]
5	hypercholesterol*[Title/Abstract]
6	essential hypercholesterolemia[MeSH Terms]
7	hypertriglycerid*[Title/Abstract]
8	hypertriglyceridemia[MeSH Terms]
9	"lipid disorder"[Title/Abstract]
10	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9
11	prevalence[Title/Abstract]
12	prevalence[MeSH Terms]
13	#11 OR #12
14	#10 AND #13

Participants/Population

This systematic review will include adults 18 years or older with dyslipidaemia (elevated total cholesterol, elevated LDL-c, elevated TG, low HDL-c).

Inclusion and exclusion criteria

We will include all cross-sectional and longitudinal observational studies reporting on the prevalence of elevated serum total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), and low high-density lipoprotein cholesterol (HDL-c) in adults above 18 years old. We will exclude studies of children or individuals with familial hypercholesterolaemia in this review. We will also exclude editorials, commentaries, reviews, letters, case series with less than 50 patients, and studies without primary data or explicit description of methods. When a study is published in two or more reports, the most comprehensive one with the largest sample size will be included.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Data extraction (selection and coding)

Two review authors will independently screen the titles and abstracts of the articles retrieved from the literature search. These citations will be imported into the Endnote software, with duplicate articles being omitted. Any studies excluded from the review will be documented with justification. Full texts of potentially eligible articles will be attained and further evaluated for final inclusion. Any disagreement among the review authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary. We will seek professional translation services for abstracts and full articles that are not available in English.

Two review authors will independently extract relevant data from individual studies using a pro forma designed specifically for this review. The following data will be extracted: first author's name, publication year, study design, country, locality (rural vs urban), setting (community or hospital-based), sample size, mean/median age, age range, proportion of men/women, any disease specific to the study population, dyslipidaemia subtypes included (ie elevated TC, elevated LDL-c, elevated TG, low HDL-c), diagnostic cut-off levels, and the number of participants with dyslipidaemia. We will consider the most frequently used cut-off for the diagnosis of dyslipidaemia. Any disagreement among the review authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary.

Risk of bias (quality) assessment

We will evaluate all included studies for quality and bias using an adapted version of the Risk of Bias Tool for Prevalence Studies developed by Hoy et al.[19] or the Joanna Briggs Institute's Checklist for Prevalence Studies[20], pending team deliberation and piloting if required.

Strategy for data synthesis

First, we will perform a narrative (descriptive) synthesis of the study results. We will assess the degree of heterogeneity in the estimates among studies as measured by the I square (I^2) statistics, which denotes the percentage of variation in estimate that is not attributable to sampling error and will be relevant only if there are more than one included study in the analysis. We adopt an I^2 level of 75% to indicate a substantial degree of heterogeneity.[21] As heterogeneity is inherent among prevalence studies, we adopt a higher threshold of I^2 estimate compared to that in a systematic review of randomized studies.

If I^2 of the pool of studies is below 75%, we will undertake meta-analysis using the random-effects model, to take into account of the inherent heterogeneity present among different prevalent studies. To address the limitations of current meta-analysis of prevalence studies, in which studies with very low

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

or very high prevalence tends to be assigned disproportionately high weight due to the marked reduction of their inferred variance towards 0, we will transform all prevalence estimates using the Freeman-Tukey transformation (arcsine square root transformation)[22] for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence interval[23] , using the MetaXL software as an add-in of Excel (EpiGear International, QLD, Australia). We will then back-transform the point estimate and 95% confidence interval, and report our results using the back-transformed figures to facilitate interpretation.

If the I^2 is 75% or above, which indicates substantial level of heterogeneity, we will explore possible contributors of heterogeneity by assessing variation between studies in terms of the following: i. Population characteristics, including geographical location, age group and sex, ii. Definition and measurement of the target condition of dyslipidaemia, including the data collection methods (e.g. self-completed questionnaire vs. interview or testing (clinical assessment and/or objective test), iii. Risk of bias of the included studies, divided by low or high risks of bias in participation and outcome measurement. If we find plausible contributors of heterogeneity, we will divide the studies into appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression (Stata 13 software).

Sensitivity analysis

If sufficient data are available, we will perform sensitivity analysis to assess the impact of excluding studies with an overall high risk of bias.

Certainty of evidence

As there is not yet an established tool to rate the certainty of evidence for prevalence studies, we will discuss our findings by incorporating our assessment of the risk of bias of included studies and our findings in sensitivity analysis.

Limitation of study

The main potential limitation of this systematic review could be the lack of studies on the prevalence of dyslipidaemia in specific regions around the world such as South America, Australia, and the Pacific Islands. In view of our limitations in time and manpower, we have opted to focus on the databases with the highest relevant yields from our experience in our search strategy, which are PubMed/MEDLINE and CENTRAL. These databases cover most relevant articles to be screened and additional databases

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

such as Web of Science and Scopus may yield little additional relevant information amidst a large body of non-relevant articles. Hence, there is a possibility of potentially missing articles with this approach.

ETHICS AND DISSEMINATION

As the current study is based on available published data, ethics approval is not a requirement. We plan to disseminate the findings from this systematic review at relevant international conferences. Then, we will submit the final report in the form of an article to a peer-reviewed journal for publication. We also plan to share the findings with the relevant health authorities such as the World Health Organization and guideline development bodies. Finally, we plan to update the review in the future as more relevant publications are produced.

Author Contributions MSMY and LNM conceived and designed the protocol. MSMY drafted the manuscript. NB, SAR, ASR, and LNM revised the manuscript for methodological and clinical content. All authors read and approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors

Conflict of interest None

Patient consent for publication Not applicable.

REFERENCES

1. Health Technology Assessment Unit. Clinical Practice Guidelines on Management of Dyslipidaemia: Ministry of Health Malaysia, 2017:105.
2. Collins R, Reith C, Emberson J, et al. Interpretation of the evidence for the efficacy and safety of statin therapy. *Lancet* 2016;388(10059):2532-61. doi: 10.1016/S0140-6736(16)31357-5 [published Online First: 2016/09/13]
3. Ference BA, Ginsberg HN, Graham I, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J* 2017;38(32):2459-72. doi: 10.1093/eurheartj/ehx144 [published Online First: 2017/04/27]
4. Silverman MG, Ference BA, Im K, et al. Association Between Lowering LDL-C and Cardiovascular Risk Reduction Among Different Therapeutic Interventions: A Systematic Review

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

and Meta-analysis. *JAMA* 2016;316(12):1289-97. doi: 10.1001/jama.2016.13985 [published Online First: 2016/09/28]

5. Emerging Risk Factors Collab., Di Angelantonio E, Sarwar N, et al. Major lipids, apolipoproteins, and risk of vascular disease. *JAMA* 2009;302(18):1993-2000. doi: 10.1001/jama.2009.1619 [published Online First: 2009/11/12]
6. Gordon DJ, Probstfield JL, Garrison RJ, et al. High-density lipoprotein cholesterol and cardiovascular disease. Four prospective American studies. *Circulation* 1989;79(1):8-15. doi: 10.1161/01.cir.79.1.8 [published Online First: 1989/01/01]
7. Hubert HB, Feinleib M, McNamara PM, et al. Obesity as an independent risk factor for cardiovascular disease: a 26-year follow-up of participants in the Framingham Heart Study. *Circulation* 1983;67(5):968-77. doi: 10.1161/01.cir.67.5.968 [published Online First: 1983/05/01]
8. Rosenson RS. Low HDL-C: a secondary target of dyslipidemia therapy. *Am J Med* 2005;118(10):1067-77. doi: 10.1016/j.amjmed.2004.12.021 [published Online First: 2005/10/01]
9. Hamazaki T, Okuyama H, Ogushi Y, et al. Towards a Paradigm Shift in Cholesterol Treatment. A Re-examination of the Cholesterol Issue in Japan. *Ann Nutr Metab* 2015;66 Suppl 4:1-116. doi: 10.1159/000381654 [published Online First: 2015/05/01]
10. Ravnskov U, de Lorgeril M, Diamond DM, et al. LDL-C does not cause cardiovascular disease: a comprehensive review of the current literature. *Expert Rev Clin Pharmacol* 2018;11(10):959-70. doi: 10.1080/17512433.2018.1519391 [published Online First: 2018/09/11]
11. Voight BF, Peloso GM, Orho-Melander M, et al. Plasma HDL cholesterol and risk of myocardial infarction: a mendelian randomisation study. *Lancet* 2012;380(9841):572-80. doi: 10.1016/S0140-6736(12)60312-2 [published Online First: 2012/05/23]
12. Arsenault BJ, Rana JS, Stroes ES, et al. Beyond low-density lipoprotein cholesterol: respective contributions of non-high-density lipoprotein cholesterol levels, triglycerides, and the total cholesterol/high-density lipoprotein cholesterol ratio to coronary heart disease risk in apparently healthy men and women. *J Am Coll Cardiol* 2009;55(1):35-41. doi: 10.1016/j.jacc.2009.07.057 [published Online First: 2010/02/02]
13. Ramjee V, Sperling LS, Jacobson TA. Non-high-density lipoprotein cholesterol versus apolipoprotein B in cardiovascular risk stratification: do the math. *J Am Coll Cardiol* 2011;58(5):457-63. doi: 10.1016/j.jacc.2011.05.009 [published Online First: 2011/07/23]
14. Robinson JG, Wang S, Smith BJ, et al. Meta-analysis of the relationship between non-high-density lipoprotein cholesterol reduction and coronary heart disease risk. *J Am Coll Cardiol* 2009;53(4):316-22. doi: 10.1016/j.jacc.2008.10.024 [published Online First: 2009/01/24]
15. World Health Organization. Global health observatory data repository Geneva: World Health Organization; 2016 [cited 2020 4 January]. Available from: https://apps.who.int/gho/data/view.main.2467?lang=en accessed 4 January 2020.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

16. Lin CF, Chang YH, Chien SC, et al. Epidemiology of Dyslipidemia in the Asia Pacific Region. *Int J Gerontol* 2018;12(1):2-6. doi: 10.1016/j.ijge.2018.02.010

17. Noubiap JJ, Bigna JJ, Nansseu JR, et al. Prevalence of dyslipidaemia among adults in Africa: a systematic review and meta-analysis. *Lancet Glob Health* 2018;6(9):e998-e1007. doi: 10.1016/s2214-109x(18)30275-4 [published Online First: 2018/08/15]

18. Baigent C, Keech A, Kearney PM, et al. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;366(9493):1267-78. doi: 10.1016/S0140-6736(05)67394-1 [published Online First: 2005/10/11]

19. Hoy D, Brooks P, Woolf A, et al. Assessing risk of bias in prevalence studies: modification of an existing tool and evidence of interrater agreement. *J Clin Epidemiol* 2012;65(9):934-9. doi: 10.1016/j.jclinepi.2011.11.014 [published Online First: 2012/06/30]

20. Munn Z, Moola S, Lisy K, et al. Methodological guidance for systematic reviews of observational epidemiological studies reporting prevalence and cumulative incidence data. *Int J Evid Based Healthc* 2015;13(3):147-53. doi: 10.1097/XEB.0000000000000054 [published Online First: 2015/09/01]

21. Barendregt JJ, Doi SA, Lee YY, et al. Meta-analysis of prevalence. *J Epidemiol Community Health* 2013;67(11):974-8. doi: 10.1136/jech-2013-203104 [published Online First: 2013/08/22]

22. Freeman MF, Tukey JW. Transformations Related to the Angular and the Square Root. *Ann Math Statist* 1950;21(4):607-11. doi: 10.1214/aoms/1177729756

23. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P 2015 Checklist

This checklist has been adapted for use from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	94
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	3–55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	249-251
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	N/A
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐	☒	N/A
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	☒	N/A
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☒	N/A
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	130-131
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	131-132
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	167-183

For peer review only

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	143-146
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	149-164
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	175-198
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	183-189
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	190-191
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	167-198
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	176-179
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	200-203
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	213-222
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	206-212
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	229-234
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	206
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	N/A
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	235-238

For peer review only

BMJ Open

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-049662.R2
Article Type:	Protocol
Date Submitted by the Author:	13-Nov-2021
Complete List of Authors:	Mohamed-Yassin, Mohamed-Syarif; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Baharudin, Noorhida; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine Abdul-Razak, Suraya; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Cardiac Vascular and Lungs Research Institute (CaVaLRI) Ramli, Anis Safura; Universiti Teknologi Mara Fakulti Perubatan, Department of Primary Care Medicine; Universiti Teknologi MARA, Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM) Lai, Nai Ming; Taylor's University, School of Medicine
Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Cardiovascular medicine, Global health
Keywords:	Lipid disorders < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY, Cardiology < INTERNAL MEDICINE

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Global prevalence of dyslipidaemia in adult populations: a systematic review protocol

Mohamed-Syarif Mohamed-Yassin[1§], Noorhida Baharudin[1], Suraya Abdul-Razak[1,2], Anis Safura Ramli[1,3], Lai Nai-Ming[4]

[1] Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Jalan Prima Selayang 7, 68100 Batu Caves, Selangor, MALAYSIA

[2] Cardiac Vascular and Lungs Research Institute (CaVaLRI), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[3] Institute of Pathology, Laboratory & Forensic Medicine (I-PPerForM), Universiti Teknologi MARA, Jalan Hospital, 47000 Sungai Buloh, Selangor, MALAYSIA

[4] School of Medicine, Taylor's University, 1, Jalan Taylor's, 47500 Subang Jaya, Selangor, MALAYSIA

[§]**Corresponding Author**:

Name: Dr Mohamed-Syarif Mohamed-Yassin

Postal Address: Department of Primary Care Medicine, Faculty of Medicine, Universiti Teknologi MARA, Selayang Campus, Jalan Prima Selayang 7, Batu Caves, 68100 Selangor, MALAYSIA.

Email: syarif8258@uitm.edu.my

Contact No.: +6012 970 2295

For peer review only

Authors' details and e-mail address:

Author 1: MSMY

Given names: Mohamed-Syarif

Family name: Mohamed-Yassin

Preferred abbreviated names: Mohamed-Yassin MS

Email: syarif8258@uitm.edu.my

Author 2: NB

Given names: Noorhida

Family name: Baharudin

Preferred abbreviated names: Baharudin N

Email: noorhida8229@uitm.edu.my

Author 3: SAR

Given names: Suraya

Family name: Abdul-Razak

Email: suraya617@uitm.edu.my

Preferred abbreviated names: Abdul-Razak S

For peer review only

Author 4: ASR

Given names: Anis Safura

Family name: Ramli

Email: anis014@uitm.edu.my

Preferred abbreviated names: Ramli AS

Author 5: LNM

Given names: Nai Ming

Family name: Lai

Email: naiming.lai@taylors.edu.my

Preferred abbreviated names: Lai NM

Word count and references:

- Abstract: 300
- Main text: 1466
- References: 23
- Table: 1

For peer review only

ABSTRACT

Introduction: Dyslipidaemia is one of the established risk factors for cardiovascular disease. Evidence from large-scale trials showed that effective treatment of dyslipidaemia can reduce all-cause and coronary mortality. To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults. We propose to perform a systematic review on the global prevalence of dyslipidaemia in adults 18 years and older.

Methods and analysis: We will identify observational studies through comprehensive literature searches. We will search: MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov. Two reviewers will independently screen the titles and abstracts, attain full text of eligible articles, extract data, and appraise the quality and bias of the included studies. Disagreement among the authors will be resolved by discussion leading to a consensus. Next, we will perform a narrative synthesis of the study results. Study heterogeneity will be assessed using I^2 statistics. If I^2 is high (≥75%), and plausible heterogeneity contributors are found, we will divide the studies into appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression. If I^2 <75%, we will undertake meta-analysis using the random-effects model and transform all prevalence estimates using the Freeman-Tukey transformation for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence. We will then back-transform the point estimate, and report our results using the back-transformed figures.

Ethics and dissemination: Ethics approval is not a requirement as this study is based on available published data. Results of this systematic review will be presented at conferences, shared with relevant health authorities, and published in a peer-reviewed journal. These results may help quantify the magnitude of dyslipidaemia globally, and guide preventative and therapeutic interventions.

Trial registration: PROSPERO registration number - CRD42020200281

ARTICLE SUMMARY

Strengths and limitations of this study

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

- This is the first systematic review on the global prevalence of dyslipidaemia in adult populations
- The overall pooled prevalence of dyslipidaemia, as well as the prevalence in different geographical regions and settings will help provide the baseline risk and absolute risk difference in studies assessing interventions for dyslipidaemia.
- Possible limitation of this systematic review could be the lack of studies and data on the subtypes of dyslipidaemia.

INTRODUCTION

Dyslipidaemia refers to lipid abnormalities consisting of either one or any combination of the following: elevated total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), or low high-density lipoprotein cholesterol (HDL-c).[1] Dyslipidaemia is one of the established risk factors for cardiovascular disease.[1] Extensive reviews had concluded that elevated LDL-c is a significant contributor to atherosclerotic cardiovascular disease (ASCVD).[2-4] Low HDL-c had also been found to be associated with CVD.[5-8] However, recent studies had questioned the role of high TC and LDL-c, in the development of atherosclerosis and CVD.[9 10] The protective role of increasing HDL-c in CVD had also been challenged[11], while some studies had shown that non-HDL-c predicts CV risk better than LDL-c.[12-14]

The World Health Organization's estimates from 2008 showed that the prevalence of hypercholesterolaemia in adults were the highest in Europe (53.7%) and America (47.7%), while South East Asia (30.3%) and Africa (23.1%) had much lower prevalence.[15] However, Lin et al. reported marked prevalence differences between different Asia Pacific countries, ranging from 9% in Indonesia to 46.9% in the Philippines.[16] For high LDL-c, high TG, and low HDL-c, the prevalence ranges from 7.8 to 47.2%, 13.9 to 38.6%, and 10.1 to 71.3%, respectively.[16] Noubiap et al. in their 2018 systematic review of dyslipidaemia in Africa, reported the prevalence of elevated TC, elevated LDL-c, elevated TG, and low HDL-c as 25.5%, 28.6%, 17%, and 37.4%, respectively.[17]

Vigorous evidence from large-scale randomised trials showed that effective treatment of dyslipidaemia can reduce all-cause mortality and coronary mortality.[18] LDL-c reduction of 1 mmol/L with statin treatment can lower the 5-year incidence of stroke, coronary revascularisation, and major coronary events by around one fifth.[2 18]

To date, there is no published systematic review on the worldwide prevalence of dyslipidaemia in adults and we aim to address this evidence gap. Hence, the objective of this systematic review is to determine the global prevalence of dyslipidaemia in adults 18 years old and older.

Here, we present the protocol for this systematic review, prepared according to the PRISMA-P 2015 guidelines.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

METHODS AND ANALYSIS

Actual start date of study

1st September 2020.

Anticipated end date of study

28th February 2022.

Patient and public involvement

No patient involved.

Study design

This will be a systematic review with meta-analysis if suitable data are available for pooling.

Search strategy

We will search PubMed/MEDLINE, Cochrane Central Register of Controlled Trials (CENTRAL) for published studies, and trial registries including the WHO International Trial Registry Platform and ClinicalTrials.gov for ongoing studies. There will be no restrictions applied on the publication period and language. The main search strategy is shown in Table 1.

Table 1 Search strategy

A. Search strategy PubMed/MEDLINE (adapted for CENTRAL)	
Search	Search terms
1	dyslipid*[Title/Abstract]
2	dyslipidemia[MeSH Terms]
3	hyperlipid*[Title/Abstract]
4	hyperlipidemia[MeSH Terms]
5	hypercholesterol*[Title/Abstract]
6	essential hypercholesterolemia[MeSH Terms]
7	hypertriglycerid*[Title/Abstract]
8	hypertriglyceridemia[MeSH Terms]
9	“lipid disorder”[Title/Abstract]
10	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9
11	prevalence[Title/Abstract]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12	prevalence[MeSH Terms]
13	#11 OR #12
14	#10 AND #13

Participants/Population

This systematic review will include adults 18 years or older with dyslipidaemia (elevated total cholesterol, elevated LDL-c, elevated TG, low HDL-c).

Inclusion and exclusion criteria

We will include all cross-sectional and longitudinal observational studies reporting on the prevalence of elevated serum total cholesterol (TC), elevated low-density lipoprotein cholesterol (LDL-c), elevated triglycerides (TG), and low high-density lipoprotein cholesterol (HDL-c) in adults above 18 years old. We will exclude studies of children or individuals with familial hypercholesterolaemia in this review. We will also exclude editorials, commentaries, reviews, letters, case series with less than 50 patients, and studies without primary data or explicit description of methods. When a study is published in two or more reports, the most comprehensive one with the largest sample size will be included.

Data extraction (selection and coding)

Two review authors will independently screen the titles and abstracts of the articles retrieved from the literature search. These citations will be imported into the Endnote software, with duplicate articles being omitted. Any studies excluded from the review will be documented with justification. Full texts of potentially eligible articles will be attained and further evaluated for final inclusion. Any disagreement among the review authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary. We will seek professional translation services for abstracts and full articles that are not available in English.

Two review authors will independently extract relevant data from individual studies using a pro forma designed specifically for this review. The following data will be extracted: first author's name, publication year, study design, country, locality (rural vs urban), setting (community or hospital-based), sample size, mean/median age, age range, proportion of men/women, any disease specific to the study population, dyslipidaemia subtypes included (ie elevated TC, elevated LDL-c, elevated TG, low HDL-c), diagnostic cut-off levels, and the number of participants with dyslipidaemia. We will consider the most frequently used cut-off for the diagnosis of dyslipidaemia. Any disagreement among the review

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

authors will be resolved by discussion leading to a consensus, with referral to a third review author if necessary.

Risk of bias (quality) assessment

We will evaluate all included studies for quality and bias using an adapted version of the Risk of Bias Tool for Prevalence Studies developed by Hoy et al.[19] or the Joanna Briggs Institute's Checklist for Prevalence Studies[20], pending team deliberation and piloting if required.

Strategy for data synthesis

First, we will perform a narrative (descriptive) synthesis of the study results. We will assess the degree of heterogeneity in the estimates among studies as measured by the I square (I^2) statistics, which denotes the percentage of variation in estimate that is not attributable to sampling error and will be relevant only if there are more than one included study in the analysis. We adopt an I^2 level of 75% to indicate a substantial degree of heterogeneity.[21] As heterogeneity is inherent among prevalence studies, we adopt a higher threshold of I^2 estimate compared to that in a systematic review of randomized studies.

If I^2 of the pool of studies is below 75%, we will undertake meta-analysis using the random-effects model, to take into account of the inherent heterogeneity present among different prevalent studies. To address the limitations of current meta-analysis of prevalence studies, in which studies with very low or very high prevalence tends to be assigned disproportionately high weight due to the marked reduction of their inferred variance towards 0, we will transform all prevalence estimates using the Freeman-Tukey transformation (arcsine square root transformation)[22] for pooling, to obtain a synthesised point estimate of prevalence with its 95% confidence interval[23] , using the MetaXL software as an add-in of Excel (EpiGear International, QLD, Australia). We will then back-transform the point estimate and 95% confidence interval, and report our results using the back-transformed figures to facilitate interpretation.

If the I^2 is 75% or above, which indicates substantial level of heterogeneity, we will explore possible contributors of heterogeneity by assessing variation between studies in terms of the following: i. Population characteristics, including geographical location, age group and sex, ii. Definition and measurement of the target condition of dyslipidaemia, including the data collection methods (e.g. self-completed questionnaire vs. interview or testing (clinical assessment and/or objective test), iii. Risk of bias of the included studies, divided by low or high risks of bias in participation and outcome measurement. If we find plausible contributors of heterogeneity, we will divide the studies into

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

appropriate subgroups for pooling of results or assess the association of plausible covariates and the prevalence estimates using meta-regression (Stata 13 software).

Sensitivity analysis

If sufficient data are available, we will perform sensitivity analysis to assess the impact of excluding studies with an overall high risk of bias.

Certainty of evidence

As there is not yet an established tool to rate the certainty of evidence for prevalence studies, we will discuss our findings by incorporating our assessment of the risk of bias of included studies and our findings in sensitivity analysis.

Limitation of study

The main potential limitation of this systematic review could be the lack of studies on the prevalence of dyslipidaemia in specific regions around the world such as South America, Australia, and the Pacific Islands. In view of our limitations in time and manpower, we have opted to focus on the databases with the highest relevant yields from our experience in our search strategy, which are PubMed/MEDLINE and CENTRAL. These databases cover most relevant articles to be screened and additional databases such as Web of Science and Scopus may yield little additional relevant information amidst a large body of non-relevant articles. Hence, there is a possibility of potentially missing articles with this approach.

ETHICS AND DISSEMINATION

As the current study is based on available published data, ethics approval is not a requirement. We plan to disseminate the findings from this systematic review at relevant international conferences. Then, we will submit the final report in the form of an article to a peer-reviewed journal for publication. We also plan to share the findings with the relevant health authorities such as the World Health Organization and guideline development bodies. Finally, we plan to update the review in the future as more relevant publications are produced.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Author Contributions MSMY and LNM conceived and designed the protocol. MSMY drafted the manuscript. NB, SAR, ASR, and LNM revised the manuscript for methodological and clinical content. All authors read and approved the final manuscript.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors

Conflict of interest None

Patient consent for publication Not applicable.

REFERENCES

1. Health Technology Assessment Unit. Clinical Practice Guidelines on Management of Dyslipidaemia: Ministry of Health Malaysia, 2017:105.
2. Collins R, Reith C, Emberson J, et al. Interpretation of the evidence for the efficacy and safety of statin therapy. *Lancet* 2016;388(10059):2532-61. doi: 10.1016/S0140-6736(16)31357-5 [published Online First: 2016/09/13]
3. Ference BA, Ginsberg HN, Graham I, et al. Low-density lipoproteins cause atherosclerotic cardiovascular disease. 1. Evidence from genetic, epidemiologic, and clinical studies. A consensus statement from the European Atherosclerosis Society Consensus Panel. *Eur Heart J* 2017;38(32):2459-72. doi: 10.1093/eurheartj/ehx144 [published Online First: 2017/04/27]
4. Silverman MG, Ference BA, Im K, et al. Association Between Lowering LDL-C and Cardiovascular Risk Reduction Among Different Therapeutic Interventions: A Systematic Review and Meta-analysis. *JAMA* 2016;316(12):1289-97. doi: 10.1001/jama.2016.13985 [published Online First: 2016/09/28]
5. Emerging Risk Factors Collab., Di Angelantonio E, Sarwar N, et al. Major lipids, apolipoproteins, and risk of vascular disease. *JAMA* 2009;302(18):1993-2000. doi: 10.1001/jama.2009.1619 [published Online First: 2009/11/12]
6. Gordon DJ, Probstfield JL, Garrison RJ, et al. High-density lipoprotein cholesterol and cardiovascular disease. Four prospective American studies. *Circulation* 1989;79(1):8-15. doi: 10.1161/01.cir.79.1.8 [published Online First: 1989/01/01]
7. Hubert HB, Feinleib M, McNamara PM, et al. Obesity as an independent risk factor for cardiovascular disease: a 26-year follow-up of participants in the Framingham Heart Study. *Circulation* 1983;67(5):968-77. doi: 10.1161/01.cir.67.5.968 [published Online First: 1983/05/01]
8. Rosenson RS. Low HDL-C: a secondary target of dyslipidemia therapy. *Am J Med* 2005;118(10):1067-77. doi: 10.1016/j.amjmed.2004.12.021 [published Online First: 2005/10/01]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

9. Hamazaki T, Okuyama H, Ogushi Y, et al. Towards a Paradigm Shift in Cholesterol Treatment. A Re-examination of the Cholesterol Issue in Japan. *Ann Nutr Metab* 2015;66 Suppl 4:1-116. doi: 10.1159/000381654 [published Online First: 2015/05/01]
10. Ravnskov U, de Lorgeril M, Diamond DM, et al. LDL-C does not cause cardiovascular disease: a comprehensive review of the current literature. *Expert Rev Clin Pharmacol* 2018;11(10):959-70. doi: 10.1080/17512433.2018.1519391 [published Online First: 2018/09/11]
11. Voight BF, Peloso GM, Orho-Melander M, et al. Plasma HDL cholesterol and risk of myocardial infarction: a mendelian randomisation study. *Lancet* 2012;380(9841):572-80. doi: 10.1016/S0140-6736(12)60312-2 [published Online First: 2012/05/23]
12. Arsenault BJ, Rana JS, Stroes ES, et al. Beyond low-density lipoprotein cholesterol: respective contributions of non-high-density lipoprotein cholesterol levels, triglycerides, and the total cholesterol/high-density lipoprotein cholesterol ratio to coronary heart disease risk in apparently healthy men and women. *J Am Coll Cardiol* 2009;55(1):35-41. doi: 10.1016/j.jacc.2009.07.057 [published Online First: 2010/02/02]
13. Ramjee V, Sperling LS, Jacobson TA. Non-high-density lipoprotein cholesterol versus apolipoprotein B in cardiovascular risk stratification: do the math. *J Am Coll Cardiol* 2011;58(5):457-63. doi: 10.1016/j.jacc.2011.05.009 [published Online First: 2011/07/23]
14. Robinson JG, Wang S, Smith BJ, et al. Meta-analysis of the relationship between non-high-density lipoprotein cholesterol reduction and coronary heart disease risk. *J Am Coll Cardiol* 2009;53(4):316-22. doi: 10.1016/j.jacc.2008.10.024 [published Online First: 2009/01/24]
15. World Health Organization. Global health observatory data repository Geneva: World Health Organization; 2016 [cited 2020 4 January]. Available from: https://apps.who.int/gho/data/view.main.2467?lang=en accessed 4 January 2020.
16. Lin CF, Chang YH, Chien SC, et al. Epidemiology of Dyslipidemia in the Asia Pacific Region. *Int J Gerontol* 2018;12(1):2-6. doi: 10.1016/j.ijge.2018.02.010
17. Noubiap JJ, Bigna JJ, Nansseu JR, et al. Prevalence of dyslipidaemia among adults in Africa: a systematic review and meta-analysis. *Lancet Glob Health* 2018;6(9):e998-e1007. doi: 10.1016/s2214-109x(18)30275-4 [published Online First: 2018/08/15]
18. Baigent C, Keech A, Kearney PM, et al. Efficacy and safety of cholesterol-lowering treatment: prospective meta-analysis of data from 90,056 participants in 14 randomised trials of statins. *Lancet* 2005;366(9493):1267-78. doi: 10.1016/S0140-6736(05)67394-1 [published Online First: 2005/10/11]
19. Hoy D, Brooks P, Woolf A, et al. Assessing risk of bias in prevalence studies: modification of an existing tool and evidence of interrater agreement. *J Clin Epidemiol* 2012;65(9):934-9. doi: 10.1016/j.jclinepi.2011.11.014 [published Online First: 2012/06/30]
20. Munn Z, Moola S, Lisy K, et al. Methodological guidance for systematic reviews of observational epidemiological studies reporting prevalence and cumulative incidence data. *Int J Evid Based*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Healthc 2015;13(3):147-53. doi: 10.1097/XEB.0000000000000054 [published Online First: 2015/09/01]

21. Barendregt JJ, Doi SA, Lee YY, et al. Meta-analysis of prevalence. *J Epidemiol Community Health* 2013;67(11):974-8. doi: 10.1136/jech-2013-203104 [published Online First: 2013/08/22]

22. Freeman MF, Tukey JW. Transformations Related to the Angular and the Square Root. *Ann Math Statist* 1950;21(4):607-11. doi: 10.1214/aoms/1177729756

23. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60. doi: 10.1136/bmj.327.7414.557 [published Online First: 2003/09/06]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P 2015 Checklist

This checklist has been adapted for use from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	N/A
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	94
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	3–55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	249-251
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	N/A
Support					
Sources	5a	Indicate sources of financial or other support for the review	☐	☒	N/A
Sponsor	5b	Provide name for the review funder and/or sponsor	☐	☒	N/A
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☐	☒	N/A
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	130-131
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	131-132
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	167-183

For peer review only

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	143-146
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	149-164
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	175-198
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	183-189
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	190-191
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	167-198
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	176-179
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	200-203
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☒	☐	213-222
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☒	☐	206-212
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☒	☐	229-234
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	206
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	N/A
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☒	☐	235-238

For peer review only